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Review of mould prediction models and their influence on mould risk evaluation

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ABSTRACT

A reliable prediction of mould risk in buildings is important to ensure a healthy environment and to avoid social and economical damage. Whereas previously the temperature ratio was often used to minimize the mould risk, nowadays– more advanced – mould prediction models can be found (e.g. isopleth systems, biohygrothermal model, ESP-r mould prediction model, empirical VTT model). These models include the main influencing factors for mould growth: the surface temperature and relative humidity. However, they are based on either experiments or assumptions and some of them neglect a third important influencing factor: the exposure time.

The current paper gives an overview of the different existing models and analyses the impact of the mould prediction model on the mould risk evaluation. To do so, the existing mould prediction models are used to predict the mould risk for different temperature and relative humidity courses. The mould risk, the time until mould growth starts and the mould intensity according to the existing prediction models are compared. Based on the obtained results, the influence of simplifications or shortcomings in the mould prediction models is discussed.

1. Introduction

Mould spores are widely disseminated in the environment. Since mould fungi enable the decomposition and recycling of dead organic material, fungi are necessary in live. In buildings however, they can deteriorate the interior finishing, which results in economical and social damage. Furthermore, mould spores can include a risk for the occupant's health due to the settling and spreading of pathogens [1–7], especially in cases of atopic individuals [8] or dangerous mould species as Aspergillus Fumigatus or Stachybotris [9,10]. Consequently, attention should be paid to correct application of all means that allow minimization of mould risk in buildings.

Before the 1970s, non-insulated buildings and severe thermal bridges resulted in high energy losses and in many mould cases. During the first energy crisis, a growing awareness of the need to improve buildings energy performance was developed. This resulted in a general 'sealing action', in which air leaks were minimized as much as possible. Few attention was however paid to the adaptation of the ventilation strategy, so an increased humidity level and increased mould growth risk was obtained. Furthermore, with the years more and more single glazed windows have been replaced by new, very tight double glazed windows. Latter renovation technique results not only in an increased air tightness but may also shift the lowest surface temperature from the windows to the outside walls. Consequently, a free detection system based on visible condensation on the windows will be lost. Another renovation technique which introduces an increased mould risk is the addition of interior insulation, since thermal bridges are difficult to avoid. Though, also new buildings can deal with mould problems due to for example built-in moisture, bad construction details inducing moisture damage or a high moisture load in combination with an unsuitable ventilation strategy [11,12].

First step in the avoidance of mould attack is the knowledge of the boundary conditions under which moulds grow. Cardinal influencing factors in the mould growth process are temperature, moisture, substrate and exposure time [8,13,14,15]. The critical value for these parameters can however differ for each mould species. Minor influencing factors are spore availability, oxygen, pH, light, roughness of the surface, salt content of the substrate, etc. [13,14].

To evaluate the mould risk at building envelopes, different design rules and prediction models can be found in literature (e.g. temperature ratio, isopleth systems, biohygrothermal model, ESP-r mould prediction model, empirical VTT-model, etc.). Each of these models includes however different simplifications or is based on different assumptions. Consequently, to obtain a reliable indication of the mould risk, the awareness of a potential influence of the model is required.

This paper compares the impact of the different existing mould prediction models on the obtained mould risk conclusions. In Section 2 the different existing mould prediction models are

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reiterated and some first remarks are drawn. In Section 3, the different mould prediction models are compared for a series of fictitious relative humidity and temperature cycles. To end, the differences between the results obtained with the different mould prediction models are discussed and the main conclusions are drawn.

2. Mould prediction models

2.1. IEA-Annex 14

To evaluate mould risk, the IEA-Annex 14 [8,16] proposes a surface relative humidity threshold dependent on the elapsed time. This relative humidity threshold is defined based on the lowest isopleth for *Aspergillus versicolor* on agar, resulting in a relative humidity threshold equal to 80%, 89% or 100% for respectively an exposure time of 1 month, 1 week and 1 day. In a next step, this proposal is simplified into a design value for the temperature ratio:

$$\tau = \frac{\theta_{s,min} - \theta_e}{\theta_i - \theta_e} \geq 0.7 \tag{1}$$

with $\theta_{s,min}$ (°C) the minimum indoor surface temperature and θ_i and θ_e respectively the inside and outside temperature (°C). A temperature ratio of 0.7 is proposed as criterion, related to an acceptable mould risk of 5%. A lower ratio introduces an unacceptable high mould risk.

Remarks: Although the temperature ratio is often used as a design criterion, an in situ study on 35 mould infested dwellings [17] indicated that this criterion may not be used as a stand-alone performance. In the research, 12 of the 35 dwellings required a temperature ratio higher than 0.7 to exclude mould growth. Though, this growth could for the 12 cases be addressed to a too low ventilation rate, rain infiltration, less heating or a too high U-value behind cupboards. It can be concluded that the criterion based on the temperature ratio is a rudimentary criterion, taking the humidity – the main influencing parameter – only indirectly into account. Besides, the design value is defined for Belgian climatic data. For other climates another design value could be preferable. Note furthermore that a relative humidity threshold of 100% in cases of 1 day exposure is questionable since liquid water hinders mould development. A threshold of 99% relative humidity is more plausible. The critical RH threshold of 80% is also recommended in [18].

2.2. Time-of-wetness

Adan [13] performed a series of experiments to study the response of the fungal volume or water content in cases of transient relative humidity courses. He observed a distinct increase in the cell volume if the relative humidity increases, which indicates that short periods of high relative humidity may not be neglected in mould analyses. To indicate the water availability under transient conditions, he introduced the time-of-wetness (TOW), given by:

$$\mathrm{TOW} = \frac{\text{cyclic wet period } (\mathrm{RH} \geq 0.8)}{\text{cyclic period } (\text{wet} + \text{dry})} \tag{2}$$

He investigated for step changes in relative humidity the effect of the low RH, the TOW and the daily frequency (f) of the high RH periods for a given TOW and concluded that a TOW below 0.5 retards mould substantially. A TOW above 0.5 results in a non-linear relation between the TOW and the mould growth and an increased mould damage. Besides, Adan developed sigmoid curves, which enable a satisfactory fit of the mould growth rate (in [13] defined as the combination of lag phase and exponential phase) of *Penicillium chrysogenum* on gypsum board material. In this, the average rating is defined as in the BS3900 standard (see Table 1).

Remarks: Due to the limited investigated substrate types and mould species, at the moment no general conclusions can be drawn based on Adan's experiments. Also the sigmoid curves, developed based on measured data on gypsum board materials inoculated with *P. chrysogenum*, cannot be used to predict fungal growth in cases of other substrates, species or environmental conditions. Furthermore, the TOW does not include a gradation in favourable relative humidity and can consequently not be used as a stand-alone performance. Besides, no criterion is indicated to the TOW-value.

2.3. Johansson's mould growth indices [19]

Johansson et al. [19] studied three mould growth indices, based on measurements on rendered façades with different thermal inertia, colour and orientation. To investigate indoor mould growth the index I_1, which expresses the fraction of time that the RH is equal to or above the threshold relative humidity RH_t, is suggested:

$$I_1 = \frac{\int_{\tau=t_0}^{t_1} f(\tau)\mathrm{d}\tau}{t_1 - t_0}, \quad f = \begin{cases} 1 \to \mathrm{RH}(\tau) \geq \mathrm{RH_t} \\ 0 \to \mathrm{RH}(\tau) < \mathrm{RH_t} \end{cases} \tag{3}$$

where $\mathrm{RH_t}$ is often set at 80%. As I_1 neglects the temperature influence, this index is not meaningful for the investigation of exterior constructions. The temperature influence is implemented in index I_2:

$$I_2 = \frac{\int_{\tau=t_0}^{t_1} f_\mathrm{T}(\tau) f_\mathrm{RH}(\tau)\mathrm{d}\tau}{t_1 - t_0} \tag{4}$$

with f_T and f_RH the functions given in Fig. 1.

To take into account a delay due to unfavourable conditions, a recovery function f_r is added to index I_2:

$$I_3 = \frac{\int_{\tau=t_0}^{t_1} f_\mathrm{r}(\tau) f_\mathrm{T}(\tau) f_\mathrm{RH}(\tau)\mathrm{d}\tau}{t_1 - t_0},$$
$$f_\mathrm{r} = \begin{cases} 0 \to f_\mathrm{T} \times f_\mathrm{RH} = 0 \text{ has been true during the last } t_\mathrm{r} \\ 1 \to \text{otherwise} \end{cases} \tag{5}$$

Remarks: Note that the function $f_\mathrm{RH}(\tau)$ as proposed by Johansson shows a retardation of the mould growth rate at very high relative humidity, which is on the average a correct approach.

The idea of I_3 is that the recovery function f_r will be 0 for a certain period after a period of unfavourable conditions. The length of the period during which the recovery function stays zero as well as the time during which $f_\mathrm{T}(\tau) \times f_\mathrm{RH}(\tau)$ should be 0 is however not mentioned. Therefore, Johansson's mould growth

Table 1
Rating scale according to coverage area [13].

Rating	Coverage area
0	No mould growth
1	Coverage ≤ 1%
2	1% < Coverage ≤ 10%
3	10% < Coverage ≤ 30%
4	30% < Coverage ≤ 70%
5	70% < Coverage

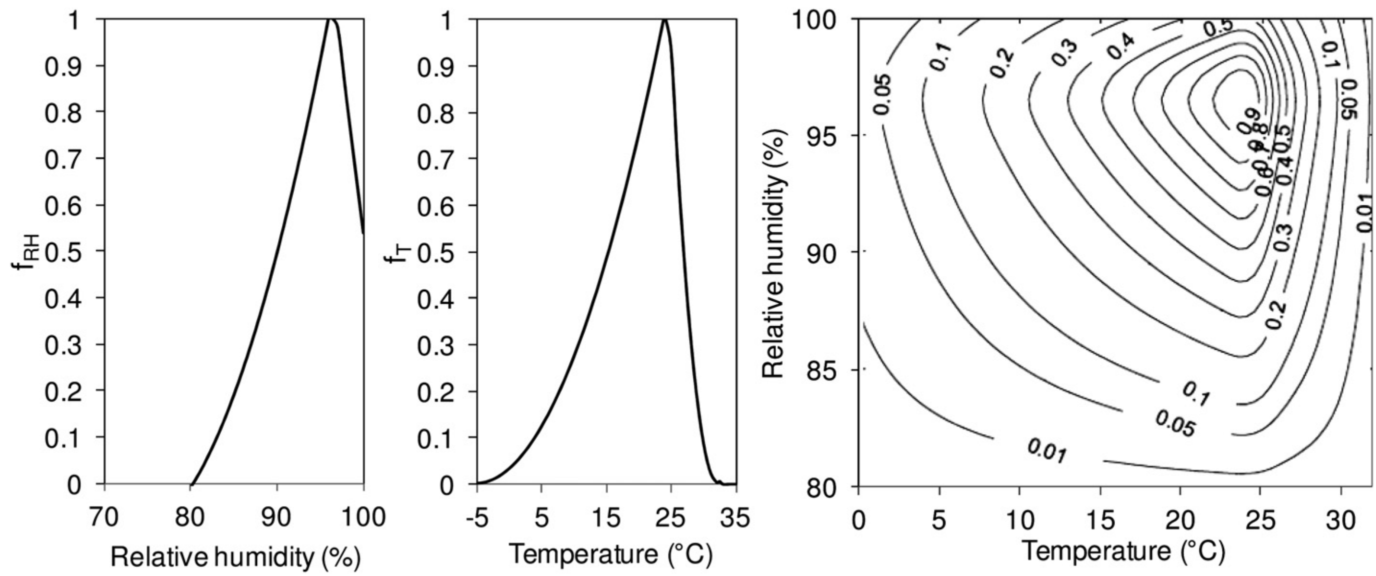

Fig. 1. f_T and f_{RH} for *Cladosporium* spp. Together with the isopleths calculated from these curves [19].

indices will not be analysed in the model comparison in this paper. Note furthermore the resemblance between index 1 and the definition of the time-of-wetness by Adan [13].

2.4. Fungal index

The fungal index, defined by Abe [20], indicates the environmental capacity for mould growth. To measure the fungal index a fungal detector in which dried fungal spores and nutrients are encapsulated, are placed in the investigated environment for a certain period. In first measurements *Eurotium herbariorum* was used as a sensor fungus. Later researches indicated that also *Aspergillus penicilliodes* or *Alternaria alternate* can be suitable as a sensor fungus for sites with respectively a lower or extremely high relative humidity [21]. In a first step of the measurement method the fungal response unit (ru) during the exposure period – which indicates the fungal growth response - is determined based on the length of the hyphae. To do so, the standard curve given in Fig. 2 is used. Latter curve is developed by substituting the fungal response unit (ru) for the incubation time (h) on the growth curve at a ratio 1:1. In a next step, the fungal index is determined by dividing the fungal response unit (ru) by the exposure period (weeks).

To make the computation of the fungal index possible, Abe [22,23] developed based on the performed measurements a database containing the fungal index values for specified temperature and relative humidity conditions. Hence, the fungal index can be used as a prediction parameter.

Remarks: Although sensor fungi different in sensitivity to relative humidity are used in the recent measurements, the standard curve seems to be only defined for the standard fungus *E. herbariorum*. Furthermore, this standard curve is obtained by incubating the fungus *E. herbariorum* under 25 °C and a relative humidity of 93.6% [24]. The reliability of this standard curve for other conditions or species is questionable.

Note also that the fungal spores in combination with nutrients are encapsulated on a plastic plate. To compare the obtained results with the mould growth sensibility on real finishing materials, further research is required.

Furthermore, environmental conditions instead of surface conditions are used in the analysis whereas surface conditions could have a larger influence on the mould growth process.

For the development of the database giving the relationship between the temperature and relative humidity conditions and the fungal index steady state conditions are used. Although a good agreement is found between measured and computed fungal indices, the reliability of this steady state approach is questionable when looking at fluctuating surface conditions. Latter conditions

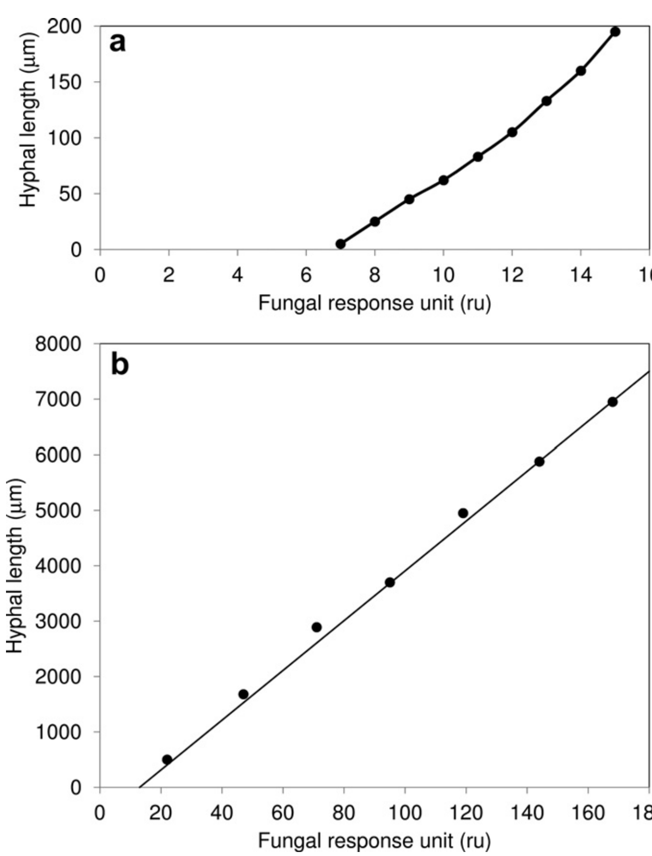

Fig. 2. Standard curves. The hyphal is defined as the distance from the spores to the hyphal tips in a, and from the edges of the spore-containing spot to the tips of the hyphae in b [22–24].

will often occur at thermal bridges, which are critical locations for mould growth.

To end, only few indications of critical fungal indices are given [20,21]. Since the database giving the relationship between the temperature and relative humidity conditions and the fungal index is not public available, the fungal index will not be analysed in the model comparison in this paper.

2.5. VTT model

The VTT model is an empirical mould prediction model developed by Hukka and Viitanen [25], in which the growth development is expressed by the mould index (M). Latter index ranges between 0 and 6 (Table 2) and can be used as a design criterion, e.g. often a mould index equal to 1 is defined as the maximum tolerable value since from that moment the germination process is assumed to start. Note that with exception of the maximum index of 6, the definition of the mould indices agrees very well with the definition for the rating used by Adan (Table 1). The VTT model is based on regression analysis of a set of measured data [26,27]. In the measurements different mixed mould species were used. The moisture content of the samples was assumed to reach equilibrium without delay. Furthermore, mould on the material was assumed not having an influence on the moisture behaviour (e.g. sorption properties) of the material [25]. Based on the different measurements, the influence of temperature, relative humidity, surface, exposure time and dry periods is included in the VTT model.

A first VTT model was based on lab experiments on pine and spruce sapwood. In this model a critical relative humidity RH_{crit} (%) can be found, defined as the lowest humidity for mould growth if the material is exposed to it for a long enough period. For wood RH_{crit} is given by

$$RH_{crit} = \begin{cases} -0.00267\theta^3 + 0.160\theta^2 - 3.13\theta + 100.0, & \text{when } \theta < 20\,^\circ\mathrm{C} \\ RH_{min}, & \text{when } \theta \geq 20\,^\circ\mathrm{C} \end{cases} \tag{6}$$

here RH_{min} is 80%. The incremental mould index in- or decrease can be calculated by use of a differential equation in which varying temperature and humidity conditions can be taken into account. For pine and spruce the incremental change in mould index is given by:

$$\frac{\mathrm{d}M}{\mathrm{d}t} = \frac{k_1 k_2}{7 \times t_{M=1}} \tag{7}$$

The factor k_1 defines the growth rate under favourable conditions and is given by:

$$k_1 = \begin{cases} 1, & \text{when } M < 1 \\ \dfrac{2}{(t_{M=3}/t_{M=1}) - 1}, & \text{when } M > 1 \end{cases} \tag{8}$$

with $t_{M=1}$ (weeks) and $t_{M=3}$ (weeks) respectively the time required to reach $M = 1$ (start growth) and $M = 3$ (first visual appearance of mould growth after the initial stage). For wood, these response times are in cases of constant temperature and humidity conditions expressed by following regression equations [25]:

$$t_{M=1} = \exp(-0.68 \times \ln(\theta) - 13.9 \times \ln(RH) + 0.14 \times W - 0.33 \times SQ + 66.02) \tag{9}$$

$$t_{M=3} = \exp(-0.74 \times \ln(\theta) - 12.72 \times \ln(RH) + 0.06 \times W + 61.50) \tag{10}$$

with W the wood specie (0 for pine, 1 for spruce), and SQ the surface quality (0 for resawn and 1 for original kiln-dried timber). To implement a moderation of the growth intensity when M approaches the maximum peak value in the range of $4 < M < 6$, the factor k_2 is included:

$$k_2 = \max[1 - \exp(2.3 \times (M - M_{max})), 0] \tag{11}$$

with M_{max} the maximum mould index (−), described by:

$$M_{max} = \max\left(1 + 7 \cdot \frac{RH_{crit} - RH}{RH_{crit} - 100} - 2\left(\frac{RH_{crit} - RH}{RH_{crit} - 100}\right)^2; 0\right) \tag{12}$$

Latter equation indicates that the critical RH is besides its dependence of the temperature also connected to the mould growth level.

A delay in mould growth caused by unfavourable conditions results in a decline of the mould index given by:

$$\frac{\mathrm{d}M}{\mathrm{d}t} = \begin{cases} -0.00133, & \text{when } t - t_1 \leq 6h \\ 0, & \text{when } 6h < t - t_1 \leq 24h \\ -0.000667, & \text{when } t - t_1 > 24h \end{cases} \tag{13}$$

with $t - t_1$ (h) the time exposed to unfavourable conditions.

Remarks: Although a delay is included in the VTT model, it should be kept in mind that Eq. (13) is developed based on a limited number of experiments and that no temperatures under 0 °C or long periods (>14 days) are investigated. One should also notice that a decrease in mould index during the unfavourable conditions

Table 2
Original mould index classification together with update based on the new experiments (bold) [27,28].

Index	Growth rate	Description		
0	No mould growth	Spores not activated		
1	Small amounts of mould on surface	Initial stages of growth	Microscopic level	
2	<10% coverage of mould on surface		Microscopic level	
3	10-30% coverage of mould on surface, **or < 50% coverage of mould (microscope)**	New spores produced	Microscopic level	Visually detectable
4	30-70% coverage of mould on surface, **or > 50% coverage of mould (microscope)**	Moderate growth	Microscopic level	Visually detectable
5	>70% coverage of mould on surface	Plenty of growth		Visually detectable
6	Very heavy, dense mould growth covers nearly 100% of the surface	Coverage around 100%		Visually detectable

does not necessarily change the visual surface appearance. Though, a delay in growth after a dry period can be clearly observed in the numerical prediction. Moreover, since the mould growth development during transient conditions is only controlled based on a daily or longer than daily basis, no experimental evidence of the hourly phenomena found in the model is available. The absence of mould decline in cases of unfavourable conditions during a period between 6 and 24 h and the restart in decline for longer periods seems physically unrealistic.

Another important remark concerns the potential influence of the time interval used in the calculation. According to Vinha [29] a shorter interval results in a larger mould index in fluctuating conditions. Vinha attributes this to the fast increase of the mould index compared to a slower decrease. Note however, that this statement is no general truth. For example, a temperature of 20 °C and cycles of 2 h 89% RH followed by 2 h 79% RH result due to the decline during the unfavourable period in a mould index of 0 when a time interval of 1 h or 2 h is used. However, when using an interval of, e.g. 4 h, 8 h, 12 h,… the averaged RH of 84% results in a mould index of 1 after ±78 days. Note also that when using for example a 6 h time interval, no decline is incalculated when unfavourable conditions occur during a period between 6 h and 24 h. Consequently, also in this case a shorter time interval (e.g. 1 h) could result in a lower mould index. When using the measurement data from a logger, the neglect of favourable or unfavourable peaks in cases of an increased preset interval for the logging time can result in respectively a smaller or larger mould index.

Main disadvantage of the VTT model is however its limitation to unpolluted spruce and pine softwood. This type of wood is widely used in Scandinavia, but hardly used for building construction elsewhere in Europe.

To end, it should be noticed that the mould growth coverage percentage which is determined on the microscopic level depends on the spore concentration used in the experiments and the magnification. The magnification used for the determination of the mould index in the VTT model is equal to 80 (Viitanen, personal communication).

2.6. Updated VTT model

Recently, the VTT model is expanded for other building materials [28,30]. The main difference with the original model is found on the microscopic level. Since for some materials already on this level a rather high mould growth coverage could be observed, the mould index classification was updated (bold in Table 2). To expand the model, the original model is updated by determining new values for the factors in the equation of dM/dt and this for the different materials. Furthermore, the RH_{min} in Eq. (6) can be changed to 85% RH for more resistant materials and SQ is set to 0 for other materials than wood.

To obtain these adaptations, large sets of laboratory experiments and monitoring of mould growth in cases of a real climate exposure [31,32] were executed on following materials: spruce board (with glued edges), concrete (maximum grain size 8 mm), aerated concrete, cellular concrete, polyurethane thermal insulation (PUR, with paper surface and with polished surface), glass wool, polyester wool, and expanded polystyrene (EPS). Since an investigation of all the building materials is not possible, four mould sensitivity classes are presented: very sensitive, sensitive, medium resistant and resistant. Table 3 gives the sensitivity classes together with the subdivision of the tested materials and an overview of the materials which belong to the different classes.

Table 4
Parameters k_1, k_2, M_{max} (A, B and C) and RH_{min} for the different mould sensitivity classes [28,30].

Sensitivity class	k_1 (if $M<1$)	k_1 (if $M\geq 1$)	M_{max} (influence on k_2)			RH_{min} (%)
			A	B	C	
Very sensitive (vs)	1	2	1	7	2	80
Sensitive (s)	0.578	0.386	0.3	6	1	80
Medium resistant (mr)	0.072	0.097	0	5	1.5	85
Resistant (r)	0.033	0.014	0	3	1	85

In the new model, the factor k_1 is determined based on the time required for a change of the mould index from $M=1$ to $M=3$. Pine sapwood is used as a reference.

$$k_1 = \begin{cases} \dfrac{t_{M=1,\text{pine}}}{t_{M=1}}, & \text{when } M<1 \\ \dfrac{2\left(t_{M=3,\text{pine}}-t_{M=1,\text{pine}}\right)}{\left(t_{M=3}-t_{M=1}\right)}, & \text{when } M>1 \end{cases} \tag{14}$$

The maximum mould index M_{max}, which influence in his turn k_2 according to Eq. (11) is given by

$$M_{max} = A + B \times \frac{RH_{crit}-RH}{RH_{crit}-100} - C\left(\frac{RH_{crit}-RH}{RH_{crit}-100}\right)^2 \tag{15}$$

k_1, k_2, A, B, C and RH_{min} for the different sensitivity classes can be found in Table 4.

The material will influence also a potential decline in mould growth [30]. Therefore, a constant relative coefficient C_{mat} is defined for the different materials. By use of this coefficient, the original decline model (Eq. (13)) can be applied in:

$$\left(\frac{dM}{dt}\right)_{mat} = C_{mat}\left(\frac{dM}{dt}\right)_{pine} \tag{16}$$

where $(dM/dt)_{mat}$ and $(dM/dt)_{pine}$ the mould decline intensity respectively for the investigated material and for pine in the original model. The coefficients C_{mat} for different materials are determined in laboratory experiments [28,30]. To model the decline of mould growth on other materials, decline classes with a factor C_{eff} are introduced, as given in Table 5.

Remarks: Although different materials are investigated for the development of the VTT model, no typical finishes or fouling are included in the research. Instead of starting from pine as a reference, typical finishing materials could be used. Moreover, question rises if the use of a reference material is advantageous. For example,

Table 3
Mould sensitivity classes [28,30].

Sensitivity class	Materials in experiment	Material groups
Very sensitive (vs)	Pine sapwood	Untreated wood; includes lots of nutrients for biological growth
Sensitive (s)	Glued wooden boards, PUR with paper surface, spruce	Planed wood; paper-coated products, wood-based boards
Medium resistant (mr)	Concrete, aerated and cellular concrete, glass wool, polyester wool	Cement or plastic based materials, mineral fibers
Resistant (r)	PUR with polished surface	Glass and metal products, materials with efficient protective compound treatments

Table 5
Decline classes and companion C_{eff}-values [28,30].

C_{eff}	Description
1	Pine in original model, short periods
0.5	Significant relevant decline
0.25	Relatively low decline
0.1	Almost no decline

four independent models – one for each sensitivity class – could perhaps result in a more reliable approach.

Furthermore, a study of external wood frame walls [29] showed some shortcomings of the model. These shortcomings are due to a lack of verification under real fluctuating conditions and temperatures below 0 °C. Real climatic conditions which consist out of long-period fluctuations, can result in a different mould decline compared to short fluctuations.

The C_{eff}-values are determined based on a limited set of experiments with relatively large scattering and should therefore be interpreted as a first approximation.

Furthermore, it is shown that the mould index when calculated under fluctuating conditions depends on the time interval.

Note that when comparing the factor $k_1 = 2$ for $M \geq 1$ in cases of the very sensitivity class (Table 4) with Eqs. (8)–(10), it seems that for 20 °C the parameters for the updated model are developed for ±97% relative humidity. For other relative humidities a small difference between the very sensitive model and the original model is obtained, as shown in Fig. 3. When using a first mould index of 1 to indicate mould risk, the start of the mould growth is however not influenced since k_1 is the same in the original and the updated very sensitive model. Note also that when comparing the k_1-values for a mould index lower and a mould index larger than 1 (Table 4), for the very sensitive and medium resistant class a larger growth rate is found when the mould index exceeds 1. On the other hand, for the sensitive and resistant class a smaller growth rate is found when the mould index exceeds 1. To end, no adaptation of the critical relative humidity for a temperature below 20 °C (Eq. (6)) is made, which results in a too low critical relative humidity for medium resistant and resistant materials if the temperature is below 20 °C.

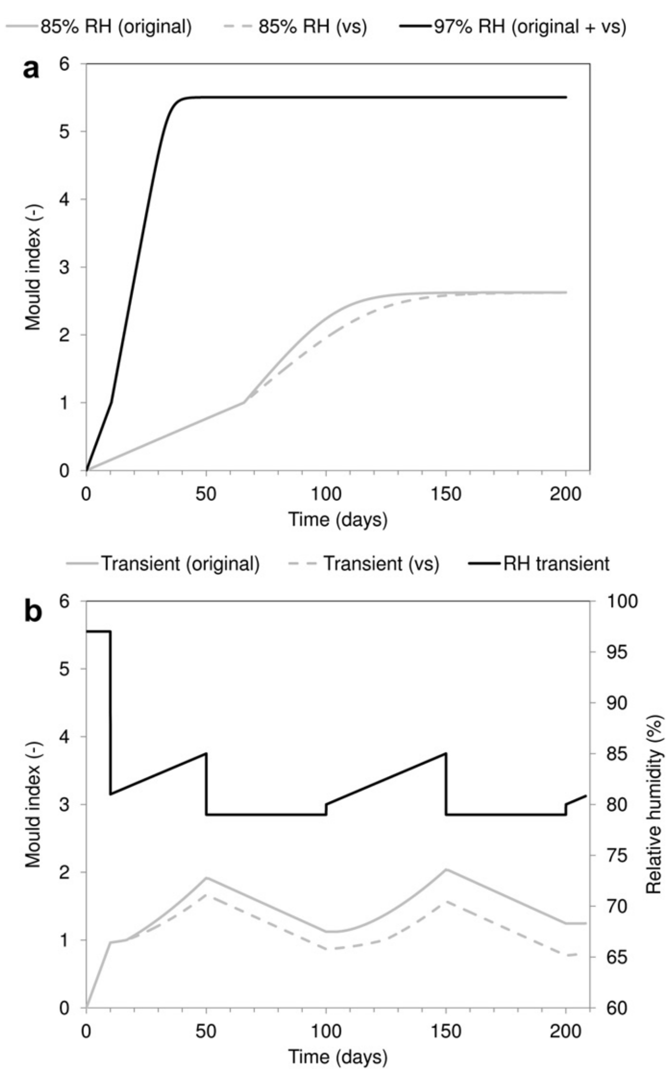

Fig. 3. Difference between the original ($W = 0$, $SQ = 0$) and updated very sensitive model (20 °C): (a) constant RH-conditions, (b) transient RH-course.

2.7. Isopleth models

Since the relative humidity (or water activity), the temperature and the exposure time are the main mould inducing factors, the relation between these factors and the mould risk is often expressed by isopleth curves. These curves separate favourable from unfavourable T- and RH-conditions for mould growth. The simplest models just provide the limit state curve, more advanced isopleth models subdivide in time till germination and growth rate. Isopleth curves are the basis in the development of mould models. Whereas the mould models described in the previous sections propose an isopleth deduced approach, other models suggest a direct use of the isopleth curves. In what follows an overview of those isopleth models can be found.

2.7.1. Ayerst [33] and Smith and Hill [34]

Ayerst [33] and Smith and Hill [34] developed isopleths for the germination times and growth rates of *Aspergillus restrictus* and *A. versicolor*. Measurements are performed on malt agar, resulting in isopleths for optimal nutrient substrates.

2.7.2. Clarke and Rowan (ESP-r model)

In the ESP-r model [35,36] the mould fungi found in buildings are subdivided into six categories with respect to relative humidity

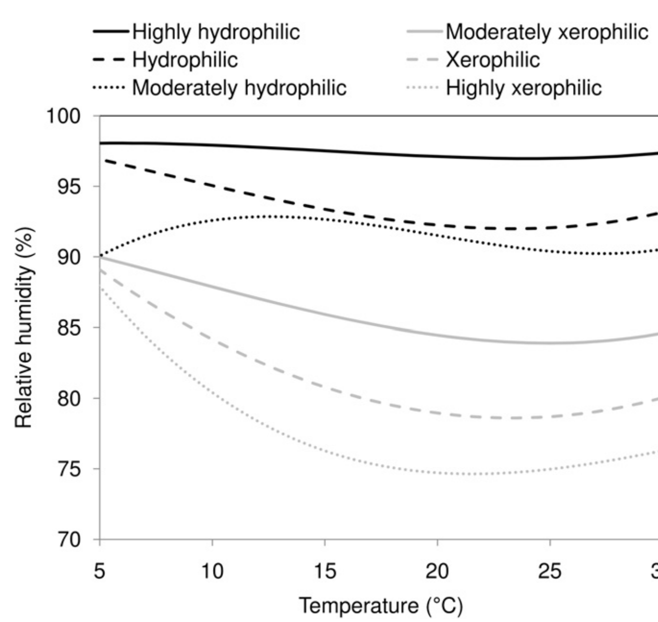

Fig. 4. Critical isopleths for the different fungi categories [35,36].

and temperature (Fig. 4). Each category is accompanied with a representative fungi:

- Highly xerophilic: *Aspergillus repens*
- Xerophilic: *A. versicolor*
- Moderately xerophilic: *P. chrysogenum*
- Moderately hydrophilic: *Cladosporium sphaerospermum*
- Hydrophilic: *Ulocladium consortiale*
- High hydrophilic: *Stachybotrys atra*

For each of these categories a growth limit curve defined by a third-order polynomial function [36] was determined based on an analysis of published data. When the relative humidity and temperature combination exceeds such a curve, mould growth of the matching fungi will occur.

Remark: Disadvantage of this model is that the exposure time is not taken into account. Consequently, since a single exceed of the isopleths is set equal to mould formation, this model assumes the worst case scenario. However, in the mould growth library of the ESP-r model the different curves are supplemented with a 'no', 'poor', 'moderate', 'good' and 'very good' growth indication. Furthermore, the isopleths are developed for an agar medium and hence overestimate the mould risk on finishing materials. Note also that the curves are developed based on experiments under constant temperature and relative humidity and that the curves are only defined above 5 °C.

2.7.3. Hens

Hens [15,17] described the isopleth of *A. versicolor* by

$$RH_{threshold} = 0.033\theta^2 - 1.5\theta + 96 \tag{17}$$

with θ the temperature (°C) and $RH_{threshold}$ the critical relative humidity (%).The equation is developed based on experiments on agar media, and defines consequently the threshold in cases of an optimal substrate. Furthermore, based on this equation the monthly mean 80% threshold is developed. For shorter periods, a logarithmic function is suggested:

$$RH_{threshold} = \min\{1, 0.8[1.25 - 0.075\ln(t)]\} \tag{18}$$

where t the time (days).

Remark: According to Hens [15,17] the choice of *A. versicolor* results from the fact that this mould specie has, compared to other moulds found on internal surfaces, the lowest isopleths and hence, highest risk. Though, when comparing the isopleths for *A. versicolor* with the isopleths for *A. restrictus* – both developed by Smith and Hill [34] – a faster growth rate is found for *A. restrictus*.

The function developed for shorter periods does not take into account the influence of the temperature. Note that a difference can be observed between the function described by Hens and the isopleths developed by Smith and Hill. For example, 82.7% relative humidity and a temperature of 20 °C during a period of 8 days results according to Smith and Hill in the start of mould growth for *A. versicolor*, whereas according to Hens during these 8 days a relative humidity of 87.5% is required to obtain mould growth.

2.7.4. Sedlbauer

Since the isopleths determined by Ayerst [33] and Smith and Hill [34] are determined for only a few fungi and moreover only for complete culture media, an expansion of these isopleths is required. Though, due to the abundance of mould species and materials, an individual isopleth system for each specie and substrate is not possible. Therefore, Sedlbauer [14] subdivided the mould species and materials found in buildings in a set of classes. In a first research, Sedlbauer [14] defined based on the health risk of the different mould species three hazardous classes:

- *Class A*: mould species which are highly pathogen and consequently not allowed to occur in buildings.
- *Class B*: mould species which are pathogen when exposed over a longer period or which cause allergic reactions.
- *Class C*: mould species which are not dangerous to health. Though, they may cause economical damage.

Based on measured minimum, optimum and maximum growth conditions for a collection of species, minimum growth conditions for the three hazardous classes were obtained. For class C only slightly different results compared to those for class B were obtained. Therefore class B and C are combined in one class B/C. The conditions below which no spore germination or growth will occur is indicated by the LIM (Lowest Isopleth for Mould)-curve. This curve is developed based on the lowest envelope of all the lowest curves of the group. After the determination of the LIM-curves, representative mould fungi for the different hazardous classes are searched. These representative mould fungi have a LIM-curve which approximates the LIM-curve described above. For class A this representative fungus is *A. versicolor*, for class B/C the mould fungi *Aspergillus amstelodami*, *Aspergillus candidus*, *Aspergillus ruber* and *Wallemia sebi* are indicated as representative fungi. Based on the growth of these representative fungi on an optimal culture medium, the isopleth systems for the hazardous classes are developed. In later researches [37] the hazardous classes are replaced by the single hazardous class K. The isopleth system for latter class includes the mould fungi, which impose according to the literature a health risk (*Aspergillus fumigatus*, *Aspergillus flavus* and *Stachybotrys chartarum*) and is defined for an optimal culture medium.

A second subdivision is made to take into account the influence of the building substrate, including possible contaminations. For the development of these curves all the mould fungi are taken into account. An overview of the different substrate categories defined by Sedlbauer is given in Table 6. As an example Fig. 5 presents the isopleth system for substrate category I. No isopleth system for substrate category III is developed, since for these substrates no mould growth will be expected. In cases of a contaminated material (soiling) of substrate category III, the isopleth system for substrate category I should be used. Furthermore, persistent materials with high open porosity belong to substrate category II. Substrate based isopleths are made in such a way that always the worst case scenario is examined, and are consequently always on the safe side in respect to preventing the formation of mould fungi.

Consequently, for the mould study on building materials following isopleth systems are obtained: LIM K (or LIM B/C and LIM A), LIM_{Mat} I and LIM_{Mat} II, each of them consisting of a germination graph and a growth rate graph.

Remark: Note that the isopleths are defined based on available measurement data of stationary laboratory experiments. This

Table 6
Sedlbauer's substrate categories [14].

Substrate category 0	Optimal culture medium
Substrate category I	Biologically recyclable building materials like wall paper, plaster cardboard, building materials made of biologically degradable raw materials, material for permanent elastic joints
Substrate category II	Biologically adverse recyclable building materials such as renderings, mineral building material, certain wood as well as insulation material not covered by I
Substrate category III	Building materials that are neither degradable nor contains nutrients

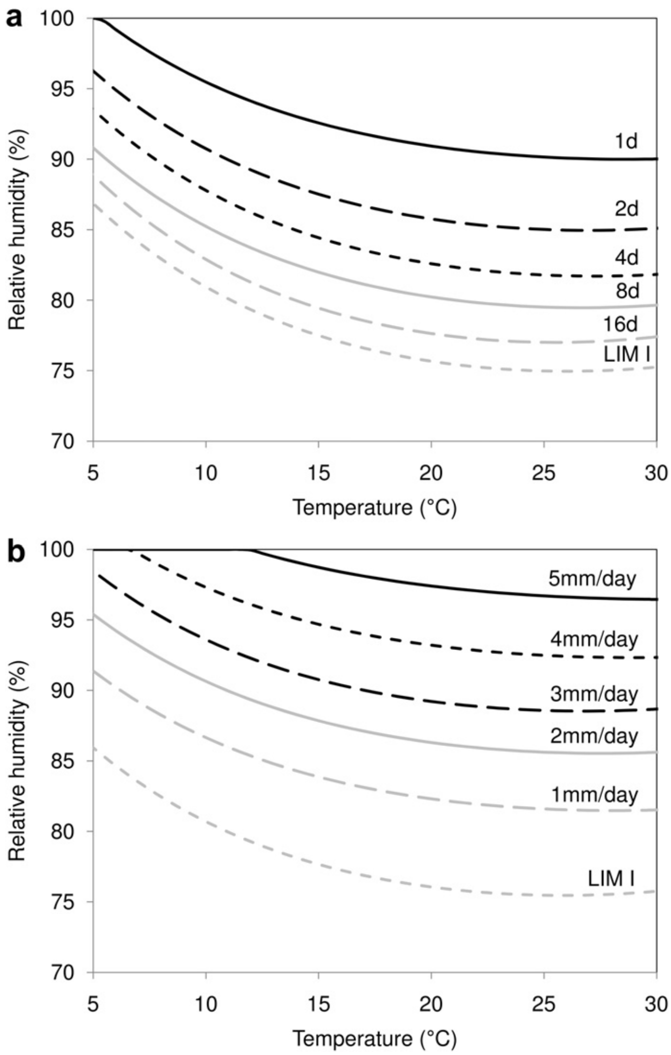

Fig. 5. Sedlbauer's isopleth system for substrate class I [38]: (a) time till germination, (b) growth rate.

limitation to stationary conditions forms the main shortcoming of the model, since an interim drying out of the spores cannot be taken into account. It seems that at very high relative humidities an extrapolation of the experimental results is made since in the isopleths no retardation in growth rate is found for these humidities. Furthermore, the growth rates (mm/day) enable only an evaluation in the figurative sense.

Besides, a straight forward use of the different curves can result in an underestimation of the mould growth. To obtain a more accurate result, an interpolation between the growth curves is required. The interpolation between the germination curves will only have a minor influence in the early stage of the mould growth since for high relative humidities only a small difference in required exposure time is needed for the germination and for lower relative humidities a negligible growth rate is obtained.

Note also that, in contrast to the findings by Hens for an optimal substrate, according to Sedlbauer's isopleth system mould growth can already start below 80% relative humidity when dealing with a substrate of category 0 or I.

2.7.5. Mould germination graph method

To take into account the temperature and relative humidity at previous time steps, Moon [39] established the mould germination graph method. In this method, each curve in the isopleth system is indicated by a certain required exposure time for initiation of mould germination. For each curve the associated accumulated exposure time can be recorded. When the accumulated exposure time for a group is equal or larger than its required exposure time, mould growth can start. Consequently, using this method also fluctuating conditions can be studied, since a delay during unfavourable conditions is implemented. The mould growth risk can be expressed as the number of mouldy risky days over a standard simulation period.

Remark: Although Moon uses the daily averaged conditions, also here a smaller time interval (e.g. an hourly calculation) is preferred. For example, when a relative humidity course equal to $RH = 70 + 20\sin((2\pi/24)t)$ with t the time (h) and a temperature of 20 °C occur, after ±274 days a mould growth of 200 mm is obtained when using hourly data. However, when performing the analysis for the daily average relative humidity (70%) no mould growth is obtained.

And although a delay during unfavourable conditions is included, a possible drying out of the spores is not considered in this model. Furthermore, when a favourable mould growth condition changes into an unfavourable condition the mould growth process is immediately switched off whereas the spore metabolism could extend the favourable growth conditions. Consequently, an underestimation of the mould growth could be obtained. On the other hand, when an unfavourable condition changes into a favourable condition, an immediately start of the mould growth process is assumed whereas possible a time lag could occur before the spore is activated. In this case an overestimation is made with the isopleth system.

2.7.6. General remark about the isopleth systems

The differences between the different isopleth systems can be subdivided into two groups: (1) the difference in the curves and subdivisions for the curves (e.g. substrate, mould species, health class) and (2) the application method (e.g. only yes/no conclusion, with or without delay). An example of a difference in the curves is shown in Fig. 6. As can be seen, the isopleths described by the model of Hens (Eq. (17)) equals the isopleth for the xerophilic species in the ESP-r model. However, although both curves are

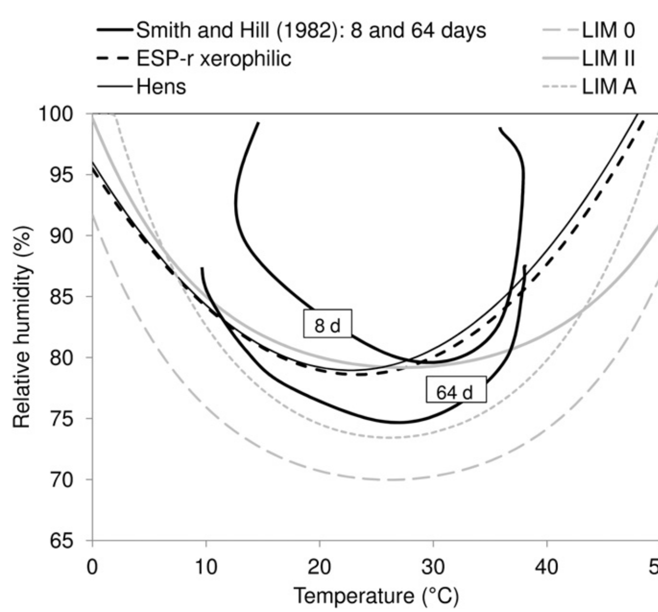

Fig. 6. Comparison of the different germination isopleths developed for *A. versicolor*, the LIM-curve for substrate category 0, substrate category II and hazardous class A.

developed for *A. versicolor*, the curves do not coincide with the isopleths developed by Smith and Hill. Note also that although Hens' isopleth and the curve for the xerophilic species in ESP-r are developed for an optimal substrate, a good agreement with Sedlbauer's LIM germination curve for substrate group II (porous materials) in cases of temperatures lower than 30 °C is found. Sedlbauer's LIM for the optimal substrate group 0 lays much lower. Furthermore, also a lower LIM-curve is obtained for health class A, although this class is represented by the A. versicolor mould specie.

2.8. Comparison between the isopleth systems and the VTT model

To make a first comparison between the VTT model and the isopleth models, based on Eq.(6) and Eq.(9) the germination isopleths can be drawn. Fig. 7a compares the isopleths for the critical relative humidity (Eq. (6)) of the original VTT model (resawn pine sapwood) with Sedlbauer's germination isopleths and with the ESP-r curve for xerophilic species. Although a very sensitive material would be rather classified as a substrate of substrate group I, the critical relative humidity according to the VTT model agrees quite well with Sedlbauer's LIM-curve for substrate category II and with the ESP-r curve for xerophilic species. Fig. 7b compares the isopleths corresponding to Eq. (9) for $t_{M=1}$ equal to 8 and 16 days with Sedlbauer's germination isopleths for 8 and 16 days. As can be observed, for a typical indoor temperature the required relative humidity to start germination is significant larger in cases of the VTT model compared to Sedlbauer's isopleth for substrate category I. Note that this difference can be due to the germination criteria used in both models. The germination criterion used in the VTT model ($M \geq 1$) seems to be less severe compared to the definition of the germination time used by Sedlbauer.

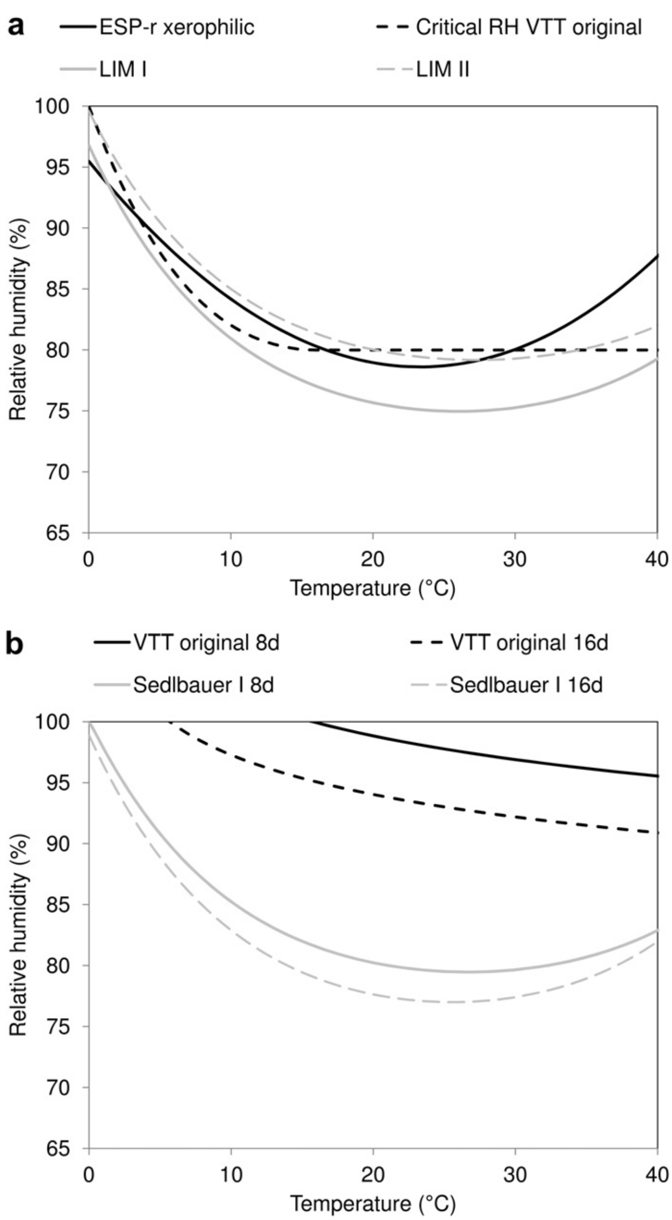

Fig. 7. Comparison between the germination time according to Sedlbauer's germination graph for substrate category I and the original VTT model: (a) critical relative humidity, (b) isopleths indicating germination after eight and sixteen days.

2.9. Biohygrothermal model

To make a more reliable prediction of the mould risk possible in cases of transient conditions, Sedlbauer extended his isopleth model with the biohygrothermal model [14,38,40,41]. In this model the moisture balance of a spore, which has a certain osmotic potential and which can consequently absorb water from the environment dependent of the transient boundary conditions, is calculated. This means also that even an interim drying out of the fungus spores can be considered. To calculate this process, the spore interior is characterised by a moisture retention curve and the spore wall by a humidity-dependent s_d-value. The spore is supposed to have germinated when a certain moisture content – the critical moisture content – is reached. From this point on metabolic processes and mould growth can start. The critical moisture content can be determined based on the moisture retention curve of the spore and the critical relative humidity found in the germination LIM-curve. The type of substrate can be taken into account by using the substrate dependent isopleth system in the determination of the critical moisture content.

By use of this model, the required time till germination and the mould growth (mm/day) can be determined. The material properties of the model spore are developed based on properties of bacteria which are slightly adapted. This adaptation is performed by fitting the material properties in this way that the calculated germination time equals the measured germination time in cases of constant conditions (which can be found in the isopleths).

Remarks: Although the simplification of the mould growth into a model spore with moisture properties makes a HAM-approach possible, care should be taken with the simplification of the rather complex mould growth process. The development of the spore properties is furthermore questionable due to the large biological difference between moulds and bacteria. Moreover, the mould species dependent isopleths developed by, e.g. Ayerst, Smith and Hill, Clarke and Rowan and Sedlbauer suggest a species dependent behaviour which is not included in the model spore.

In the biohygrothermal model an initial relative humidity is given to the spore. In WUFI-Bio the default value for this relative humidity is set at 50%. Though, according to WUFI-Bio [42], this value may be chosen between 40% and 80% and will influence only the beginning of the process. However, this means that the germination time depends on the initial spore relative humidity. Consequently, the fit of the spore properties will be an approximation. Note also that when starting from a spore relative humidity of 80%, the germination can start immediately since the critical moisture content is determined based on the LIM Isopleth, which lays lower than 80% for substrate category 0 and I. Table 7 compares for different surface relative humidities the time till germination according to the germination isopleths and to WUFI-Bio in case of

Table 7
Comparison of the time till germination according to the germination isopleths and WUFI-Bio and acquired initial spore relative humidity to obtain the germination time given by the germination isopleths for substrate I.

Surface relative humidity (%)	Germination time (h)		Initial spore RH to obtain the same germination time (%)
	Germination isopleths	WUFI-Bio (initial spore RH = 50%)	
90.94	24	13	33
85.75	48	44	48
82.58	96	126	62.5
80.23	192	265	70.4
77.63	384	817	76.9

substrate category I and an initial spore relative humidity of 50%. As can be seen, with the biohygrothermal model a shorter germination time is obtained for high surface relative humidities, which can be due to the use of the LIM-curve for the determination of the critical moisture content whereas in the isopleth model the germination curves are used. On the other hand, for low surface relative humidities a larger time till germination is found with the biohygrothermal model. Probably, this is caused by the time required to obtain the surface relative humidity in the spore. When starting from an initial spore relative humidity of 50% this 'delay' seems too long. Table 7 gives also for the different surface relative humidities the initial spore relative humidity which should be used to obtain the same germination time with the isopleth system and the biohygrothermal model. As shown, a lower or higher initial spore relative humidity could be used to compensate respectively the under- or overestimation obtained with the biohygrothermal model. This is of course no longer possible in case of transient conditions where the spore relative humidity will be obtained based on the conditions during the process.

2.10. Relationship between the biohygrothermal model and the VTT model

The mould growth in millimeters per day is a reasonable unit to describe the length of the mycel at the beginning of the growth process. In a later growth stage, however, a meshwork of mycels differing in area and thickness is obtained. From this point, the calculated growth, which can reach very large values, can only be used as a comparative factor [43]. Therefore, Krus et al. [43] developed a conversion function to transform the calculated growth, in millimeters, into the mould index. Based on simulations of the mould growth risk with both the VTT and the biohygrothermal model, the relation between the mould index and mould growth in millimeters is investigated. By limiting the biohygrothermal calculation till the moment that a mould index of 6 is obtained with the VTT model and by including reductions in growth according to the VTT model, the relation shown in Fig. 8 is obtained. A transfer function in the form of a BET function is fitted.

Remark: Since in the VTT model and the biohygrothermal model different substrate categories are defined, first a comparison between both definitions should be made. Table 8 gives, based on Table 3 and Table 6, the relation assumed between the sensitivity classes used in the VTT model and the substrate categories defined by Sedlbauer.

Although the conversion function mentioned above is developed based on several simulations, a large difference between the VTT mould index and the WUFI-Bio mould index can be obtained, as shown in Fig. 9. As can be seen, compared to the VTT mould index no moderation of the growth towards the maximal mould index is found for the WUFI-Bio index and a maximum mould index of 6 is obtained. Note especially the large difference for the short

Fig. 8. Relationship between the mould index and the mould growth [43].

exposure to high relative humidity (3 h 97%–21 h 75%; 1 d 97%–6 d 75%). According to the WUFI-Bio mould index after 170 days a moderate growth is predicted (M > 4), whereas when using the VTT mould index no mould growth is predicted (M < 1). Furthermore, the slow changes (1 d 97%–6 d 75%, 7 d 97%–23 d 75%) show clearly the absence of decline in mould index in the WUFI-Bio model. It can be concluded that the BET function in Fig. 8 does not give a good relationship between the VTT mould index and the mould growth for each relative humidity (and temperature) course.

2.11. General remark about the mould risk criteria

In previous paragraphs the different models were given. Note however, that also different definitions for the start of mould growth can be found for the different models. In the VTT model a mould index equal to 1 is defined as the start of the germination process and consequently as the value above which a mould risk exists. Note however that another critical mould index could be defined by the user of the mould model. For example, if visible mould growth is defined as criterion a critical mould index equal to 3 can be pressed forward. In the biohygrothermal model a spore moisture content higher than the critical moisture content indicates the start of germination. However, to evaluate the mould risk, in the latter model the mould growth per year is used to evaluate the mould risk [42]. To do so, a 'signal light' defines the risk:

- Mould growth > 200 mm/year: 'red light', not acceptable.
- 50 mm/year < Mould growth < 200 mm/year: 'yellow light': additional evaluation is necessary.
- Mould growth < 50 mm/year: 'green light': usually acceptable

However, no information is found on the basis the mould growth limits have been defined on.

Table 8
Relation assumed between the sensitivity classes used in the VTT-model and the substrate categories used by Sedlbauer.

VTT sensitivity class	Sedlbauer's substrate category
	0
Very sensitive (vs)	I
Sensitive (s)	I
Medium resistant (mr)	II
Resistant (r)	III

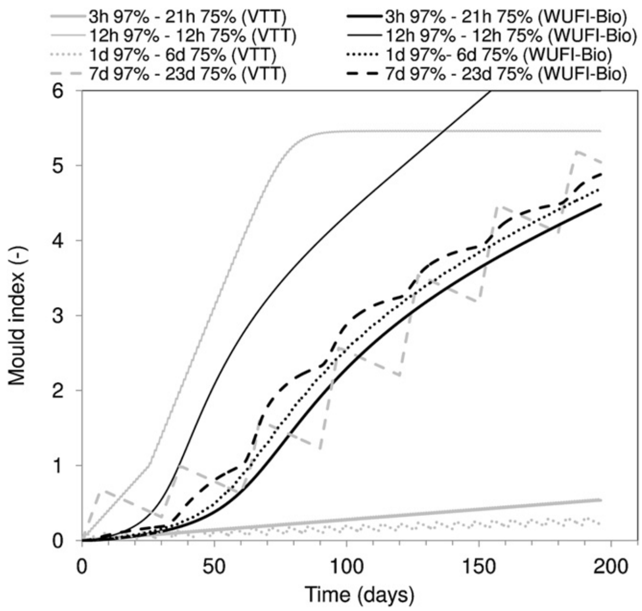

Fig. 9. Comparison of the VTT mould index (original model, $W = 0$, $SQ = 0$) and WUFI-Bio mould index (category I) for 20 °C.

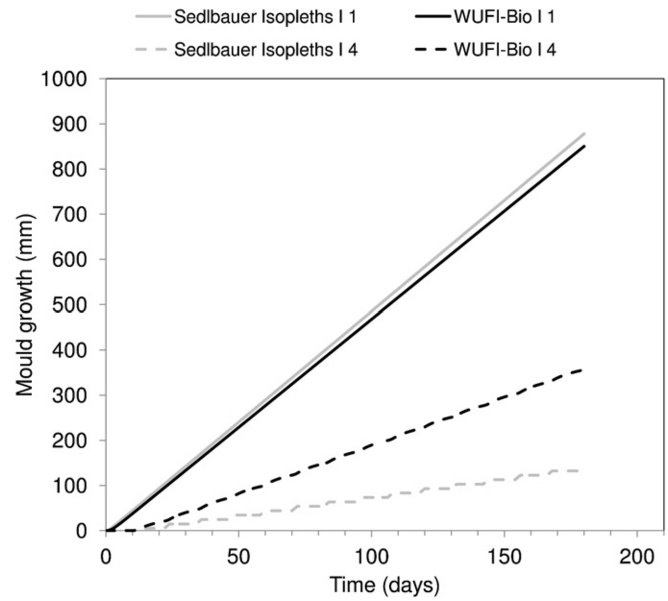

Fig. 10. Mould growth in function of time. The conditions are given in Table 9.

3. Comparison of the mould prediction models

To investigate the differences between the existing mould prediction models, five fictitious stepwise relative humidity and temperature conditions (Table 9) are assumed. The analysis is subdivided into three steps: the investigation of (1) the mould intensity in function of time, (2) the predicted mould risk in combination with the time after which a potential risk starts and (3) the total mould growth intensity after one year. Note however that the scope of the mould prediction models is to evaluate the mould risk and not to determine the exact mould growth intensity or the starting point. Therefore, the obtained start times and intensities may not be interpreted too precisely and are only used to analyse the working principle of the different models and its potential impact on the mould risk conclusions.

To determine the mould growth with the biohygrothermal model and the WUFI-Bio mould index the WUFI-Bio software [42] is used. For the initial spore relative humidity the default value of 50% is assumed. When using the isopleth systems a linear interpolation between the growth curves is made. For both models, substrate category I is assumed. In the original VTT model pine wood is assumed as substrate ($W = 0$, $SQ = 0$), while in the updated VTT model the sensitive class s is used (see Table 8 for the relation between the sensitivity classes and Sedlbauer's substrate categories). To evaluate the mould risk based on Sedlbauer's isopleth system in cases of transient conditions Moon's germination graph method is used. An identical approach is used with the short time isopleth developed by Hens. The calculations are performed based on a 1 h time interval.

3.1. Mould intensity in function of time

To discuss the mould intensity in function of time, in this paragraph the course of the mould growth and of the mould indices obtained for *condition 1* and *condition 4* is shown.

3.1.1. Mould growth

Since the spore properties in the biohygrothermal model are fitted based on Sedlbauer's isopleth system, for constant conditions identical results should be obtained when using both models. However, as shown in Fig. 10, for *condition 1* the mould growth rate obtained with the isopleth system is slightly higher compared to the growth obtained with the biohygrothermal model. Reason for this can be found in the mould growth at the beginning. The mould growth starts at a later time when using the isopleth system than when using the biohygrothermal model since in the biohygrothermal model the LIM-curve is used for the start of the germination. Though, when using the biohygrothermal model a smaller mould growth rate is observed at the start of the mould growth process which nullifies the 'profit' made in the beginning. Note however, that this start of the mould growth process is also influenced by the initial spore relative humidity as explained earlier.

For *condition 4* the average mould growth rate obtained with the biohygrothermal model is shown to be almost twice the mould growth rate obtained with Moon's germination graph method. The reason for this is twofold. At first, the germination process starts earlier for the biohygrothermal model. Though, this has again only a minor impact due to the smaller growth rate in the beginning. The second and main reason for the higher mould growth is the prolongation of the favourable spore conditions – and consequently of the mould growth process – during periods of

Table 9
Analysed conditions.

	Time 1	Relative humidity 1 (%)	Temperature 1 (°C)	Time 2	Relative humidity 2 (%)	Temperature 2 (°C)
1	Constant	97	20			
2	6 h	97	20	6 h	58	20
3	10 h	58	20	2 h	97	20
4	10 d	58	20	2 d	97	20
5	6 h	97	20	6 h	97	10

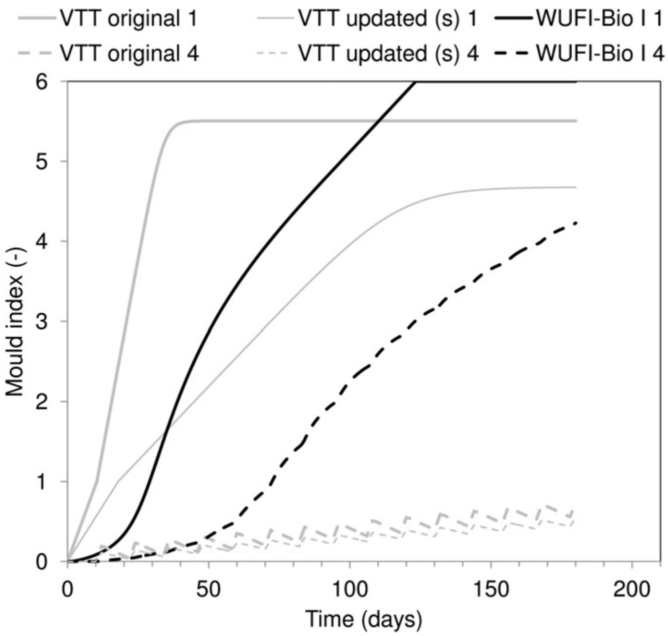

Fig. 11. VTT and WUFI-Bio mould index in function of time. The conditions are given in Table 9.

unfavourable conditions when using the biohygrothermal model. When using Moon's germination graph method an immediately pause of the mould process is assumed.

3.1.2. Mould indices

Fig. 11 compares the VTT mould index (both original and updated model) and the WUFI-Bio mould index in function of time for the different conditions. For *condition* 1, the increase of the WUFI-Bio mould index in the beginning is found to be smaller compared to the increase found for the VTT mould indices. After some time, a steeper increase is found for the WUFI-Bio mould index and the WUFI-Bio mould index reaches a value between the mould indices obtained with the original VTT model and the updated sensitive model. Whereas the VTT mould index stops at a maximal mould index, the WUFI-Bio mould index increases until a value of 6 is obtained. For *condition 4*, a large difference is found between the WUFI-Bio mould index and the VTT mould indices. Whereas the VTT mould

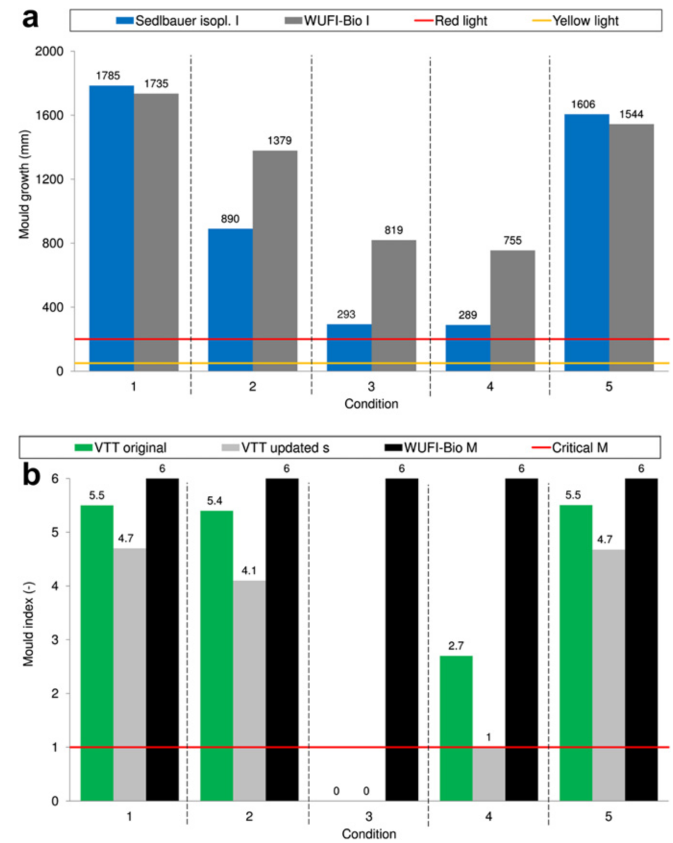

Fig. 13. Mould growth intensity after 1 year: (a) mould growth, (b) mould index. The conditions are given in Table 9.

indices are smaller than 1 (no mould risk) after 180 days, a WUFI-Bio mould index larger than 4 (moderate growth) is found.

3.2. Time till germination and mould risk

An overview of the time until germination occurs according to the different models can be found in Fig. 12. Since the existing models are mainly developed based on steady-state conditions, for constant conditions (as condition 1 of Table 9) a reliable and similar

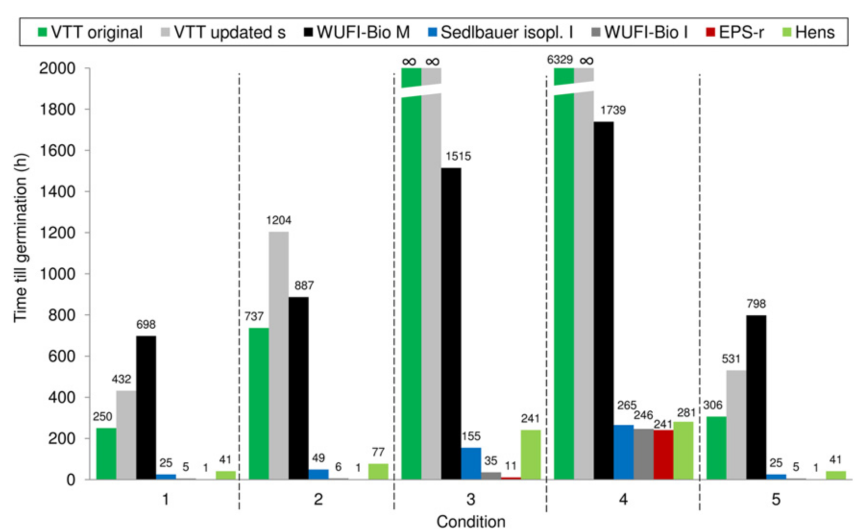

Fig. 12. Time till germination. The conditions are given in Table 9. The results given for the ESP-r model are not valid for highly hydrophilic species for which no mould risk occurs.

Table 10
Overview of the different mould prediction methods: (1) temperature ratio, (2) time-of-wetness, (3) original VTT-model, (4) updated VTT-model, (5) ESP-r isopleth model (Clarke and Rowan), (6) isopleth curve Hens, (7) isopleth system Sedlbauer, (8) Moon's mould germination graph method, (9) biohygrothermal model.

	1	2	3	4	5	6	7	8	9
T influence included?	Yes	No	Yes	Yes	Yes	Yes (long exposure time) No (short exposure time)	Yes	Yes	Yes
Growth at $T < 0$ °C?	Yes		No	No	No	Yes	No	No	Slight growth
RH influence included?	Indirectly	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Substrate taken into account?	No	No	Yes (Wood)	Yes	No	No	Yes	Yes	Yes
Estimation of growth (G) or only indication of start (S)?	S	S	G	G	S	S	G	S	G
Growth delay possible?			Yes	Yes			No	Yes (but no drying out)	No (zero growth)
Limit growth value?			Yes	Yes			No		No
Corresponding simulation tool			TCCC2D or Latenite	TCCC2D or Latenite	ESP-r				WUFI-Bio

prediction would be expected with the different models. However, for *condition 1* a large difference in time till germination is found. Note furthermore that for the investigated conditions the shortest time till germination is obtained by the ESP-r model since in this model a single exceed of the critical conditions is sufficient to indicate a mould risk. However, when a highly hydrophilic mould specie would be chosen for the analysis no mould risk would be obtained. A small increase of the time till germination is found for the biohygrothermal model (when using the '$w \geq w_{crit}$'-criterion), Moon's germination graph method (Sedlbauer's isopleth system) and Hens' isopleth for short time exposures. Moreover, when those models are arranged based on the time till germination, the same sequence is found for the investigated conditions. Besides, the time till germination obtained with these models is equal for *condition 1* and *condition 5*. In cases of the biohygrothermal model this identical time is obtained since the germination process starts already before the temperature changes in *condition 5*. In Moon's germination graph method the identical time is obtained since as well the periods of 97% RH and 20 °C as the periods of 97% RH and 10 °C exceeds the 1-day germination curve. In cases of Hens' isopleth for short exposure times no temperature influence is included.

When using the VTT mould index or the WUFI-Bio mould index in combination with the '$M \geq 1$'-criterion a much larger time till germination is found. Note furthermore, that for latter models no general sequence is found.

3.3. Mould intensity

An overview of the mould intensity after one year obtained with the different models can be found in Fig. 13. As can be observed in Fig. 13a, more mould growth is obtained with the biohygrothermal model compared to the results obtained with Moon's germination graph method (Sedlbauer's isopleth system), except for *condition 1* and *condition 5*. The reason for the less mould growth in case of *condition* 1 is already mentioned in Section 3.1.1. For *condition 5*, the lower value is probably due to the assumption of an isothermal spore in the biohygrothermal model, which results in the absence of a prolongation of the favourable spore conditions. All the investigated conditions result according to the 'signal light'-rule in an unacceptable mould growth. A comparison of the mould indices (Fig. 13b) shows that for all five investigated courses a maximum WUFI-Bio mould index of 6 is obtained. Reasons for this are the absence of a decline in this model and the unreliable BET-transformation. Note furthermore that the VTT mould indices obtained for *condition 3* are lower than the critical value of 1. Consequently, for this case no mould risk is predicted whereas the other models will predict a mould risk. When using the mould indices an equal intensity is found for *condition 1* and *condition 5*, since for these cases the maximal mould indices are obtained.

4. Conclusion

Since mould growth can have an impact on the indoor air quality and the occupant's health and since it can introduce economical and social damage, attention should be paid to mould growth prevention. To predict the mould growth risk one of the several mould prediction models found in literature can be used. These models can be subdivided in rather basic prediction parameters (time-of-wetness, Johansson's indices) and more in depth deterministic models (VTT model, isopleth systems, biohygrothermal model). However, basic models as well as more advanced mould models make a simplification of the very complex mould process. Consequently, care should be taken when analysing the results.

In a first part of this paper an overview of the different existing mould prediction models was given. It was shown that, although a lot of research has already been carried out to come to a more accurate and reliable prediction, still some shortcomings, physically unrealistic phenomena, contradictions, etc. can be found in the current available models.

In a second part, it was shown that, due to simplifications or assumptions made in the different models, a different conclusion may be drawn in the analysis depending on the model used. To investigate the effect of the different models the mould risk, the start of the germination process and the mould intensity were predicted for fictitious conditions. It was shown that for some conditions no mould risk is predicted with the VTT model whereas according to the other models mould risk exists. Furthermore, a large spread was found for the germination times obtained with the different models. The largest germination times were found for the mould indices (VTT, WUFI-Bio), which can be due to the germination criterion used in this model ($M \geq 1$). Also for the mould intensity some contradictions are found between the models. For example, in cases of cycles of 10 h 58% RH followed by 20 h 97% RH and a temperature of 20 °C an original and sensitive VTT mould index lower than 1 were obtained, whereas the WUFI-Bio mould index reached a value of 6. The course of the WUFI-Bio mould index in function of time was furthermore shown to differ also from the VTT mould index for the other conditions. Apparently, the BET-function given by Krus et al. [43] seems not reliable for every condition. Besides, no decline during unfavourable periods was found in the WUFI-Bio mould index. Latter absence could also be noticed in the biohygrothermal model and Moon's germination graph method. Consequently, also for these models a large mould growth was predicted. When comparing the biohygrothermal model with Moon's germination graph method (based on Sedlbauer's isopleths) small till larger differences were found. For the steady state condition, more mould growth was obtained with the biohygrothermal model compared to Moon's germination graph method. On the other hand, for the transient relative humidities an opposite result was found. Reason for this is the prolongation of the favourable spore conditions during unfavourable periods. However, for transient temperature conditions less mould growth was obtained with the biohygrothermal model since the assumption of an isothermal spore results in the absence of a prolongation of the favourable spore conditions. Furthermore it should be noticed that when using Moon's germination graph method – or Sedlbauer's isopleth system in cases of constant conditions – an interpolation of the growth curves is preferable, which makes the method less user friendly. In this investigation a linear interpolation between the growth curves was assumed. Another interpolation can result in slightly different intensities. Though, the general conclusions will remain.

The reader should furthermore keep in mind that the difference between the prediction methods can be caused by the difference in isopleth curves, the working principle of the model (only yes/no conclusion, with/without delay, use of a fictitious spore, etc.) or the mould risk criterion ($M \geq 1$, signal light, exceed of the critical moisture content, etc.).

Since the temperature ratio is a performance indicator of a building detail, a comparison between this ratio and the other prediction models was not possible based on the fictive conditions. A comparison of the mould risk for two thermal bridges based on the mould prediction models, including the temperature ratio criterion, can be found in [44].

To end, Table 10 gives a concise overview of the different mould prediction models. Though, it should be noticed that a more in depth study of especially the mould growth under transient conditions is still necessary to be able to define the most reliable prediction model and to solve current contradictions between the models. To do so, additional measurements in lab conditions as well as in situ are required.

Furthermore, it must be noticed that until now all mould models are deterministic models. A further improvement could be to develop prediction models which include a spread in germination time and growth rate. Besides, a variation in material properties, moisture load, transfer coefficients, etc. or bad workmanship can result in mould growth in contradiction to a theoretical absence of mould growth in cases of a deterministic approach.

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Nomenclature

Symbols

C	decline coefficient, –
f	daily frequency, /day
M	mould index, –
mr	medium resistnant
r	resistant
RH	relative humidity, %
s	sensitive
SQ	surface quality
t	time, h or day
vs	very sensitive
W	wood species

Greek symbols

τ	temperature ratio, –
θ	temperature, °C

Subscripts

crit	critical
e	outside
i	inside
mat	material
max	maximum
min	minimum
r	recovery
s	surface
T	temperature

Abbreviations

A.	Aspergillus
IEA	International Energy Agency
LIM	Lowest isopleth for mould
TOW	time-of-wetness

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